

Welcome to Code Crunch!

C5 | CRUNCH LABS: AI & Data Science

[Mon 1-2PM, Wed 12:30-1:30PM]
Weekly | Spring 2025

Attendance

Earn a Certificate of
Completion by attending
at least 70% of the
workshops



CODECRUNCH.CS.FIU





Unit #7

Ensemble Models

C5|Crunch Labs: Crunch AI Data Science

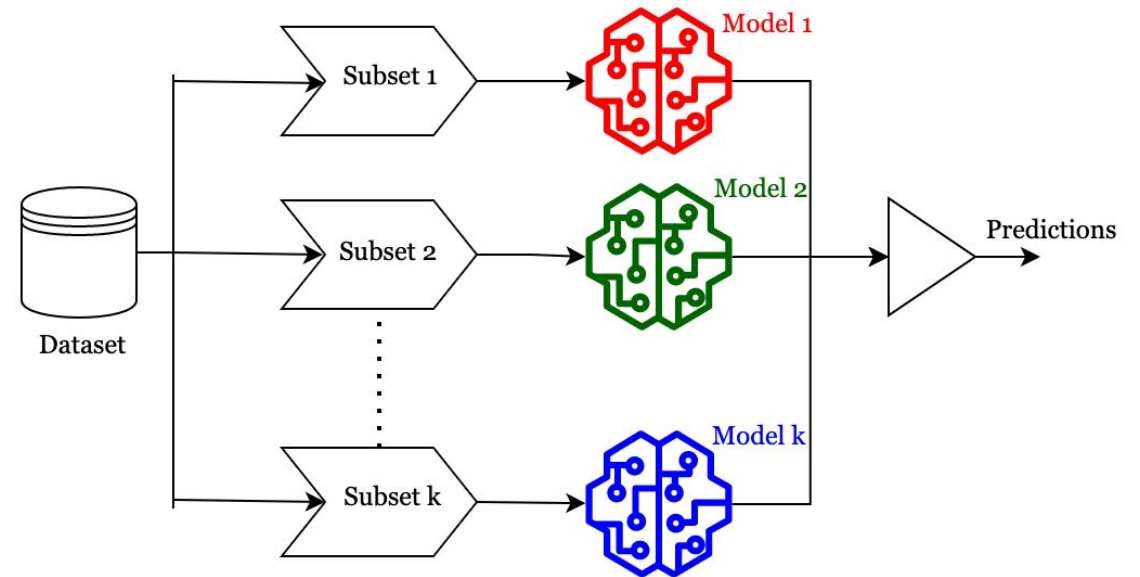
Ensemble (aan-saam-bl)

What is ensemble Learning?

- a machine learning technique that combines multiple models (often called base learners or weak learning) to create a more powerful and accurate predictive model
- The goal is to improve performance, reduce overfitting, and make the model more robust compared to using a single model

Why is ensemble Learning?

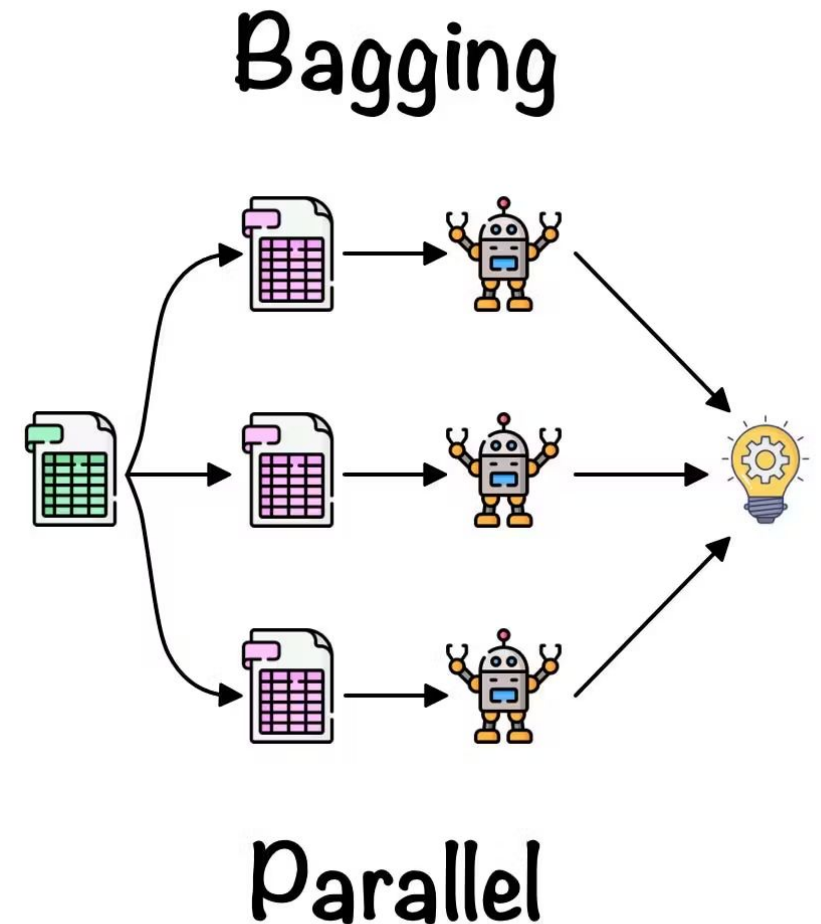
- Increased Accuracy
- Reduced Overfitting
- Better Generalization
- Error Reduction



Types of Ensemble models

Bagging

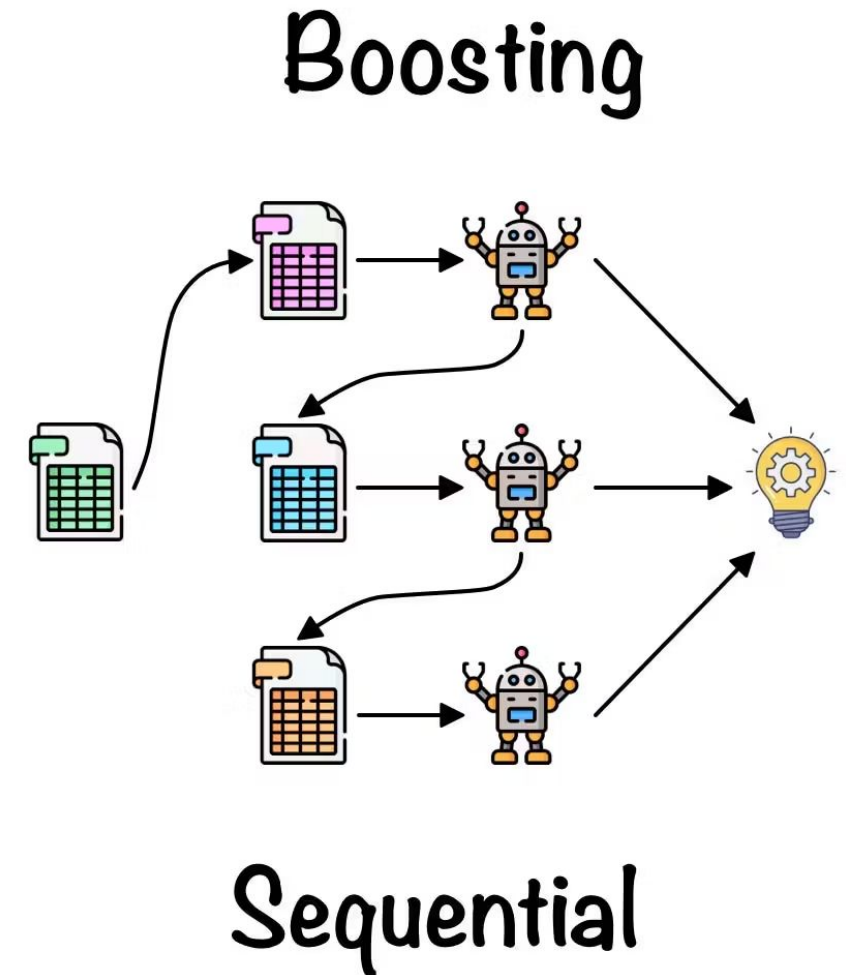
- Multiple models are trained independently on different random subsets of the data
- depending on the type of tasks there is a few ways to carry out the final decision
 - averaging the outputs (for regression)
 - majority voting (for classification)
- Examples:
 - Random Forest (an ensemble of decision trees)



Cont.

Boosting

- Models are trained sequentially, where each model focuses on correcting the errors of the previous model
- more weight is given to difficult-to-predict instances
- Examples:
 - AdaBoost
 - XGBoost
 - Gradient Boosting

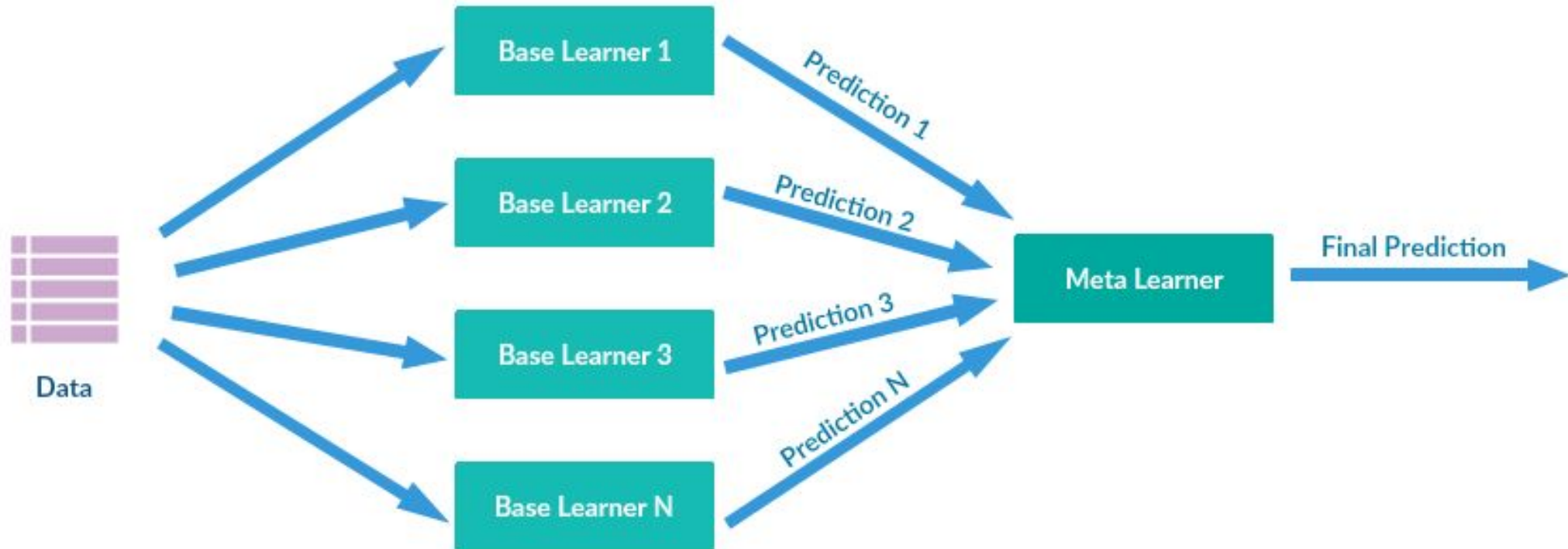


Cont. Cont.

Stacking

- Multiple models (base learners) make predictions, and their outputs are used as input for a meta-model
- The metal-model learns how to best combine the base models' predictions
- Examples:
 - combining SVM, Decision Trees, and SVM as base models with a Random Forest as the meta-model

Cont. Cont.





Q&A Session



Attendance



linktr.ee/CODE.CRUNCH



Join us for the next sessions



Thursdays: 11AM – 12PM & 2PM – 3PM

Friday: 3PM – 4PM

Monday: 12PM – 1PM



RSVP lu.ma/CODECRUNCH



Your feedback helps us improve. Share your thoughts!

Go to our website: go.fiu.edu/codecrunch